

data_curve_fit

June 23, 2025

```
[1]: import matplotlib.pyplot as plt
import numpy as np
from astropy.io import fits
from read_L1642_data import *
from scipy.optimize import minimize
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[2]: #read L1642 data
#dat = read_L1642_data(band='u')
def chisqrt(x, arg):
    band = arg
    dat = read_L1642_data(band=band)

    # Wavelengths
    # u,      b,      y,      385,      415
    # 3500.0, 4670.0, 5550.0, 3840.0, 4160.0
    Aobs      = dat.Aobs
    Aobs_err  = dat.Aobs_err
    Iobs      = dat.Iobs
    Iobs_err  = np.amax([dat.Iobs_err1, dat.Iobs_err2], axis=0)
    #Iobs_err = (dat.Iobs_err1 + dat.Iobs_err2)/2.0
    #Iobs_err = np.sqrt(dat.Iobs_err1**2 + dat.Iobs_err2**2)/2.0

    #w      = np.where(Aobs > 0.5)[0]
    #Aobs    = Aobs[w]
    #Aobs_err = Aobs_err[w]
    #Iobs    = Iobs[w]
    #Iobs_err = Iobs_err[w]
    Aobs    = Aobs[:-2]
    Aobs_err = Aobs_err[:-2]
    Iobs    = Iobs[:-2]
    Iobs_err = Iobs_err[:-2]

    I0  = x[0]
    C1  = x[1]
    C2  = x[2]
    tau = Aobs / 1.086
    Imod = I0 * np.exp(-C1*tau)*(1.0-np.exp(-C2*tau))
    chi2 = np.sum((Iobs - Imod)**2/Iobs_err**2) / (Iobs.size - 3.0)
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    return chi2

def plot_model_out(band):
    method = 'Nelder-Mead'
    #method = 'Powell'
    x0      = [61.0, 0.1, 0.5]
    bounds = ((0.0, None), (0.0, 1.0), (0.0, 1.0))
    result = minimize(chisqrt, x0, args = (band), bounds=bounds, method=method)
    #result = minimize(chisqrt, x0, args = (band), method=method)
    x      = result.x
    chi2    = result.fun

    I0     = x[0]
    C1     = x[1]
    C2     = x[2]

    xmax   = 1.0/C2*np.log(1.0+C2/C1)
    ymax   = I0 * (C1/(C1+C2))*(C1/C2)*C2/(C1+C2)
    print(x, chi2, result.success)
    print(xmax, ymax)

    tau    = np.arange(0.0, 15.0, 0.1)
    Aext   = tau * 1.086
    Imod   = I0 * np.exp(-C1*tau)*(1.0-np.exp(-C2*tau))
    dat    = read_L1642_data(band=band)

    lab    = r'$I=%4.1f\, e^{-%4.3f\tau}(1-e^{-%4.2f\tau})$' % (I0, C1, C2)
    plt.plot(Aext, Imod, label=lab)
    plt.errorbar(dat.Aobs, dat.Iobs, xerr=[dat.Aobs_err], yerr=[dat.Iobs_err1,
↳ dat.Iobs_err2], fmt='o', markersize=4.0, color='black')

    plt.legend(frameon=False)
    plt.xlim(0.0, np.amax(dat.Aobs)*1.1)
    plt.ylim(0.0, np.amax(dat.Iobs)*1.1)
    plt.xlabel(r'$A_{\rm %s}$' % band)
    plt.ylabel(r'$I_{\rm %s}$' % band)

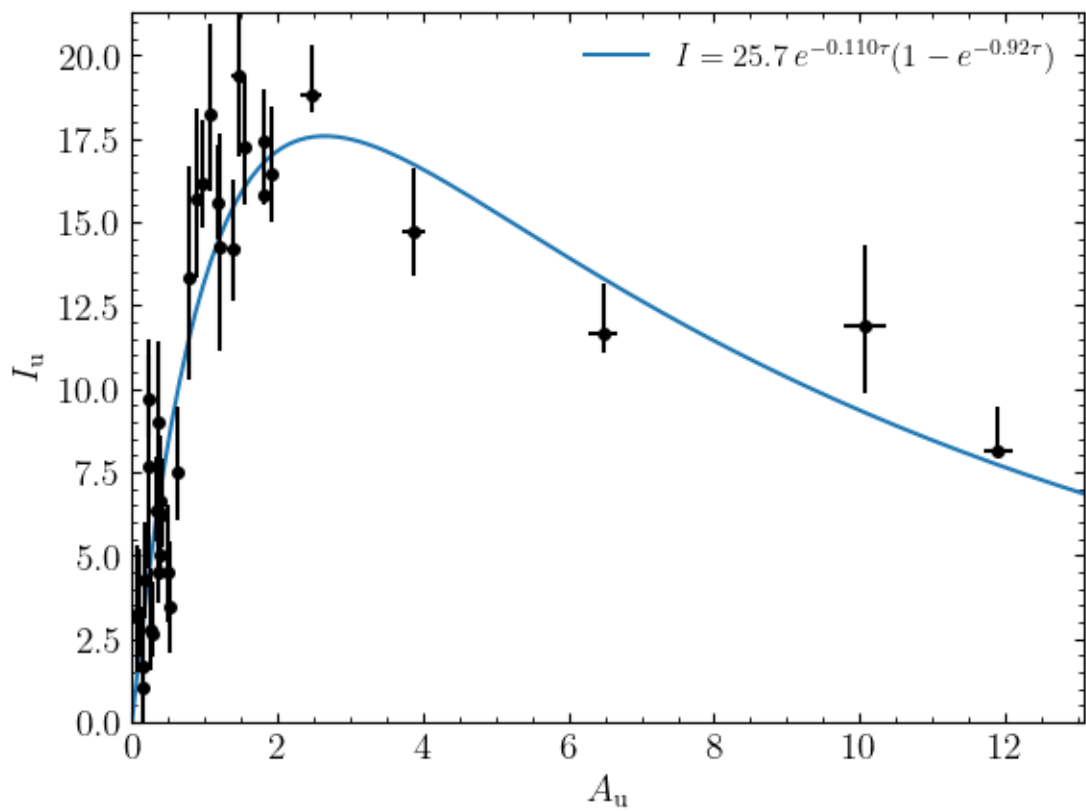
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[3]: plot_model_out('u')
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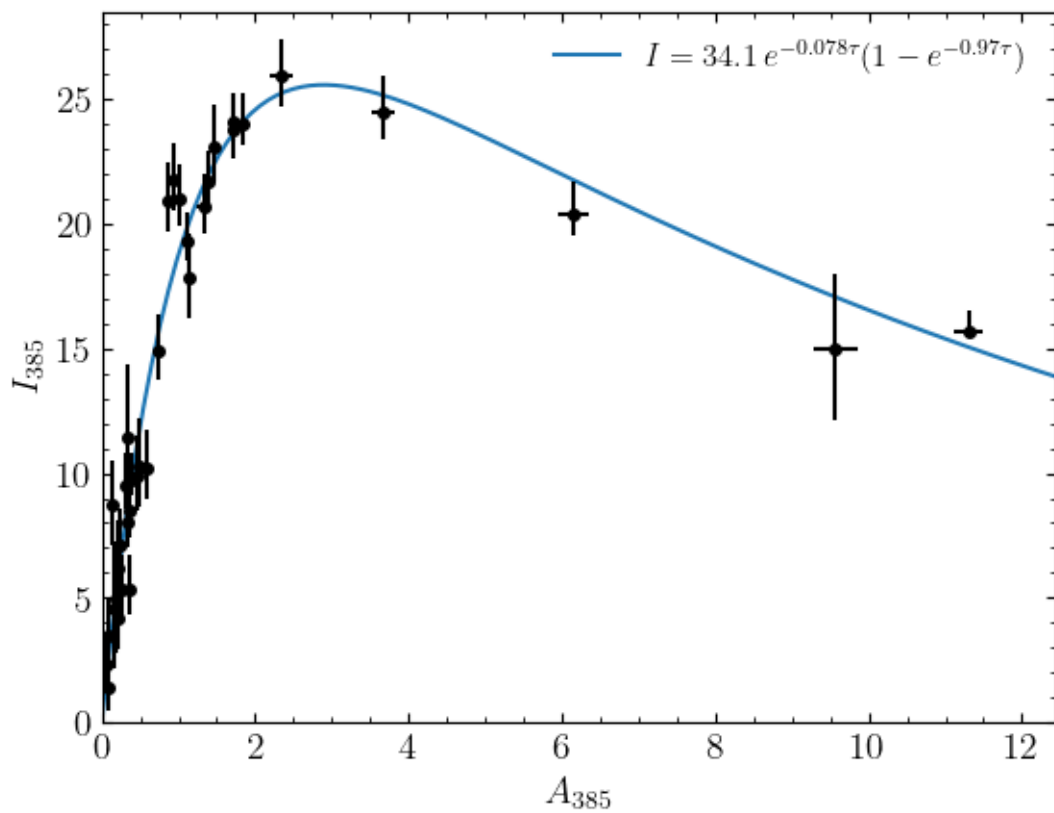
[25.71130255  0.10974397  0.91852707] 1.365601374068905 True
2.4359477823566533 17.57956977722917

```



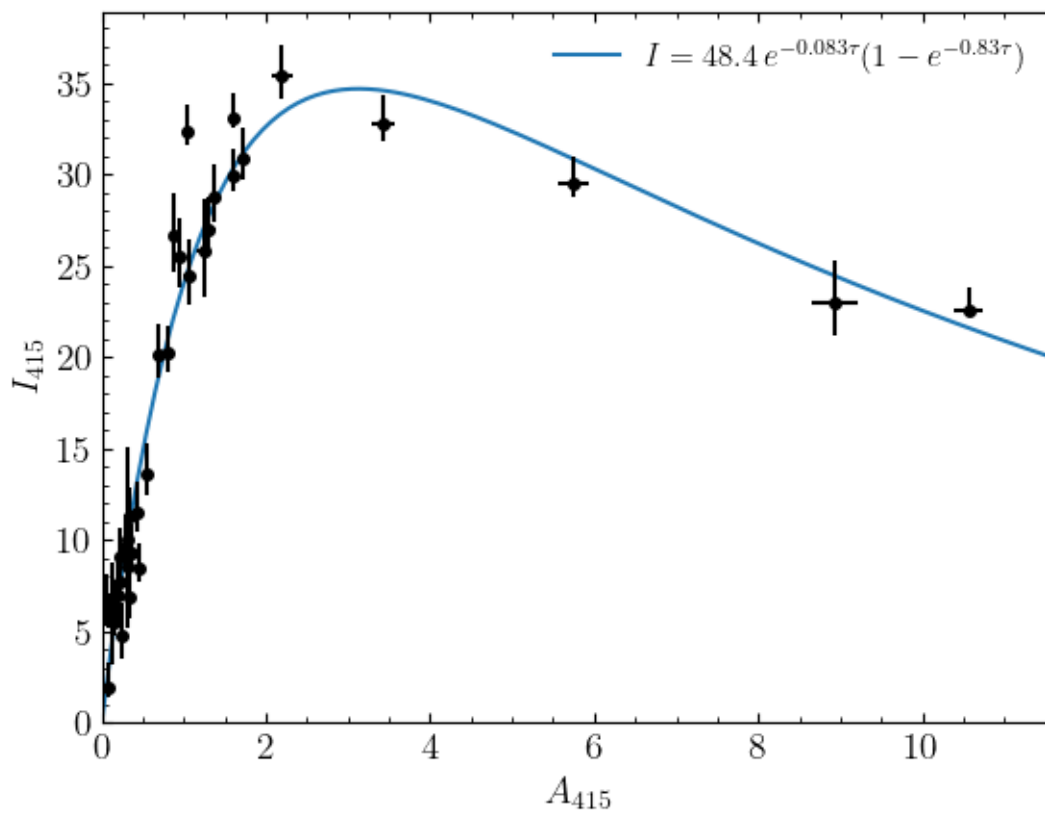
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[4]: plot_model_out('385')
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[34.05792262  0.07842455  0.97443483] 1.5265788602246573 True
2.665265959995382 25.575454705643672
```



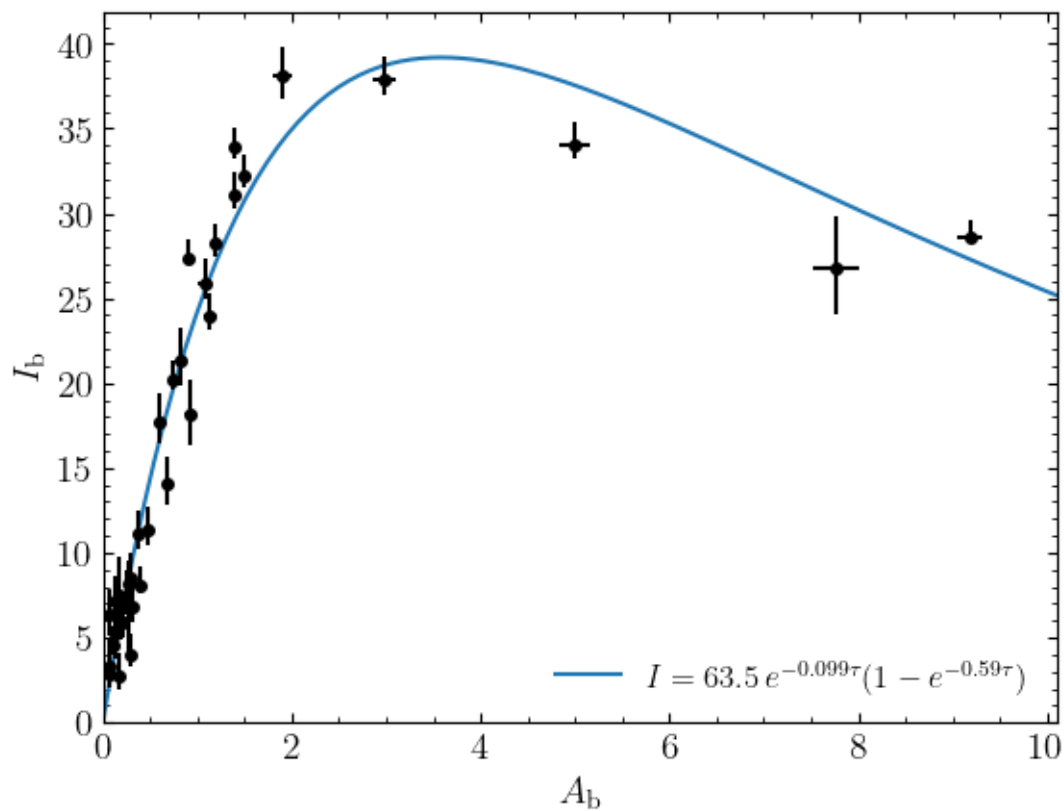
```
[5]: plot_model_out('415')
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[48.43769433  0.08301938  0.83376557] 1.411821437715351 True
2.8806644775179393 34.68150981822245
```



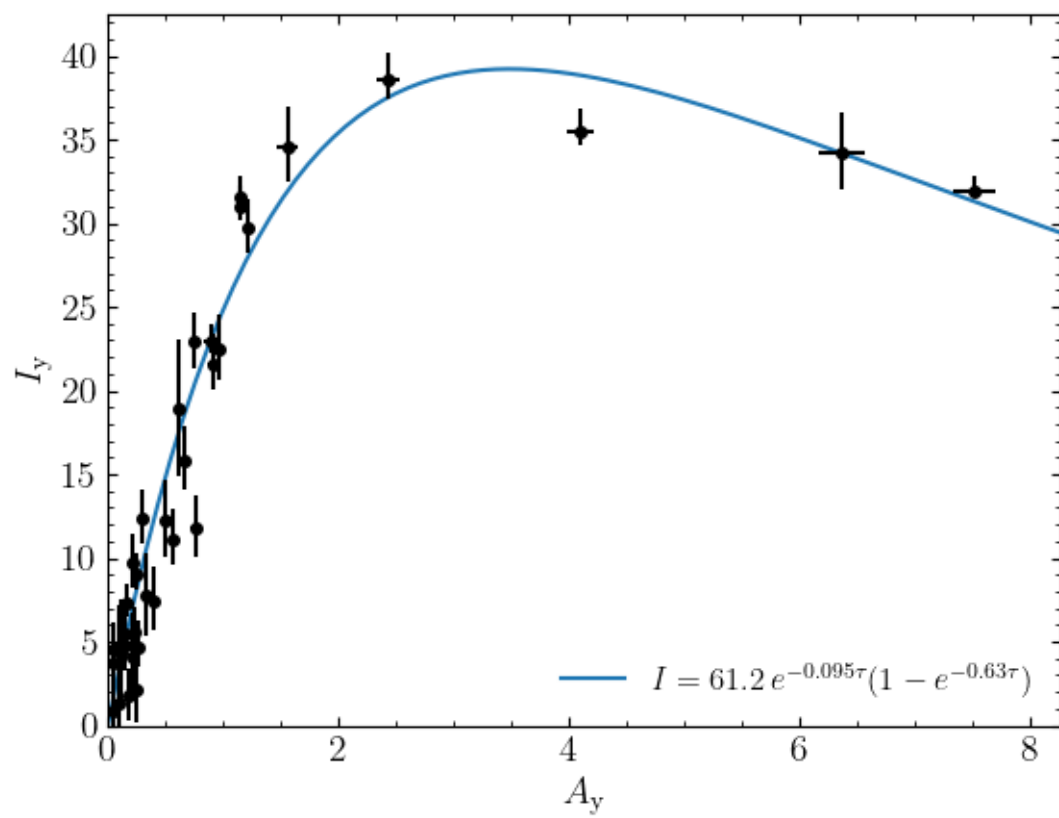
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[6]: plot_model_out('b')
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[63.51711194  0.09906544  0.58665095] 2.97622137085267 True
3.2978443055849462 39.19592785871823
```



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[7]: plot_model_out('y')
```

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[61.18255438  0.09509057  0.63477746] 3.523306104555518 True
3.2106274179534977 39.21159259851401
```



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